

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V) AI PRACTICAL

Practical –II

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Aim a) Create methods add(), multiply(), subtract(), divide() with suitable parameters and call these methods using concept of C# delegate.

Algorithm:-

1. Start the program.
2. Declare a delegate **Operation** and create a **Calculator** class with **Add()**, **Multiply()**, **Subtract()**, and **Divide()** methods.
3. Create an object of the **Calculator** class in the **Main()** method.
Declare a delegate object and assign each calculator method to it one by one.
4. Invoke the delegate to perform addition, multiplication, subtraction, and division, and display the results.
5. Display the student's name and roll number, then stop the program.

Code :-

```
using System;
delegate void Operation(int a, int b);
class Calculator
{
    public void Add(int a, int b)
    {
        Console.WriteLine("Addition=" + (a + b));
    }
    public void Multiply(int a, int b)
    {
        Console.WriteLine("Multiply=" + (a * b));
    }
    public void subtract(int a, int b)
    {
        Console.WriteLine("subtract=" + (a - b));
    }
    public void divide(int a, int b)
    {
        Console.WriteLine("divide=" + (a / b));
    }
}
class program
{
    public static void Main(string[] args)
```

```

{
    Calculator c = new Calculator();
    Operation obj;
    obj = c.Add;
    obj(20, 10);
    obj = c.Multiply;
    obj(20, 10);
    obj = c.subtract;
    obj(20, 10);

    obj = c.divide;
    obj(20, 10);
    Console.WriteLine("Achal yadav T055");
}
}

```

Output:-

```

Addition=30
Multiply=200
subtract=10
divide=2
Achal yadav T055

```

Aim b) Write a program using multicast delegate.

Algorithm:

- 1; Start the program.
- 2; Declare a delegate `Message` and create a class `Msg_Print` with methods `Hello()`, `Msg()`, and `Thank()`.
- 3; Create an object of the `Msg_Print` class in the `Main()` method.
- 4; Create a delegate object and assign the `Hello()` method to it.
- 5; Add the `Msg()` and `Thank()` methods to the same delegate using the `+=` operator.
- 6; Invoke the delegate to display all three messages, then stop the program.

Code:-

```

using System;
delegate void Message();
class Msg_Print
{
    public void Hello()
    {
        Console.WriteLine("Hello To C# COOING");
    }
    public void Msg()
    {
        Console.WriteLine("This is your Welcome Message");
    }
    public void Thank()

```

```

    {
        Console.WriteLine("Thank you for Coding");
    }
}
public class Program
{
    static void Main(string[] args)
    {
        Msg_Print mp = new Msg_Print();
        Message msg = mp.Hello;
        msg += mp.Msg;
        msg += mp.Thank;
        msg();
    }
}

```

Output:-

```

Hello To C# COOING
This is your Welcome Message
Thank you for Coding

```

c) Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

Algorithm:

1. Start the program.
2. Create a **BankAccount** class with **AccountNumber** and a **Balance** property.
3. Use the **set** accessor to allow only non-negative balance values; otherwise, display an error message.
4. Create an object of the **BankAccount** class in the **Main()** method.
5. Assign the account number and balance, then display them and attempt to set a negative balance.
6. Display the student's name and roll number, then stop the program.

Code:-

```

using System;
using System.Dynamic;

```

```

class BankAccount

```

```

{
    public int AccountNumber;

    private double balance { get; set; }
    public double Balance
    {
        get
        {
            return balance;
        }
        set
        {
            if (value >= 0)
            {
                balance = value;
            }
            else
            {
                Console.WriteLine("Balance Cannot Be Negative.");
            }
        }
    }
}

class Program
{
    public static void Main(string[] args)
    {
        BankAccount account = new BankAccount();
        account.AccountNumber = 1334;
        account.Balance = 100;

        Console.WriteLine("Account Number: " + account.AccountNumber);
        Console.WriteLine("Balance: " + account.Balance);

        account.Balance = -1000;
        Console.WriteLine("Achal Yadav T055");
    }
}

```

Output:-

Account Number: 1334

Balance: 100

Balance Cannot Be Negative.

Achal Yadav T055